

Adversarial Analysis

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1. Purpose

This document presents a good-faith adversarial analysis of both the evaluation/monitoring service a

2. Attack on Claim Conclusions

2.1 C1: Cancellation Reduction

****Claim:**** System reduced day-of-surgery cancellations by 23%

****Adversarial Arguments:****

1. ****Missing Endpoint:**** The `cancel\_rate` field cannot be mapped to day-of-surgery cancellation. An
2. ****Unknown Treatment:**** The treated services are not identified. An adversary can argue any estima
3. ****Co-intervention:**** The scheduling policy change co-occurred. An adversary can argue the entire
4. ****Proxy Assumption:**** The all-Site A proxy is not justified. Site A has 12 services; if only 2-3

****Defense (Weak):**** The analysis uses sensitivity analysis to show bounds. However, this doesn't val

2.2 C2: Readiness Time Reduction

****Claim:**** System reduced median referral-to-readiness by 4.5 days

****Adversarial Arguments:****

1. ****Same Identification Failure:**** Same treatment assignment problem as C1.
2. ****Competing Risk:**** Cancellations are a competing risk. An adversary can argue the analysis incor
3. ****Administrative Censoring:**** The study ends at 2026-07-31. An adversary can argue late cases are
4. ****Wrong Time Zero:**** Using referral time vs. assessment time. An adversary can argue time zero sh

****Defense (Weak):**** Marginal structural models attempt to address censoring, but assumptions are unt

2.3 C3: Clinician Agreement

****Claim:**** Clinicians agreed in 89% of cases

****Adversarial Arguments:****

1. ****Label Selection:**** The 500 labeled pairs are not random. An adversary can argue they over-repre

2. ****No Adjudication:**** Two reviewers disagree. An adversary can argue we don't know who is right.
 3. ****LLM Conflict:**** The LLM judge shares the model family. An agreement with the system is circular
 4. ****Wrong Population:**** The labeled sample is not the full displayed recommendation population.
- **Defense (None):**** The analysis explicitly states the claim is unsupported with these data.

3. Attack on Evaluation Service

3.1 Drift Detection

****Vulnerability:**** PSI thresholds are arbitrary

An adversary can argue:

- 0.10/0.25 thresholds are not validated
- Different thresholds would give different results
- The monitor could miss gradual drift

3.2 Calibration

****Vulnerability:**** ECE depends on bin count

An adversary can argue:

- 10 bins is arbitrary
- Results change with different binning
- The metric doesn't capture all calibration issues

3.3 Alert Budget

****Vulnerability:**** Budget could mask real problems

An adversary can argue:

- Capping alerts at 3/week could miss real issues
- Ranking by "expected harm" is subjective
- The budget is arbitrary

3.4 Provenance

****Vulnerability:**** Self-reported commit

An adversary can argue:

- The service reports its own commit
- No external verification
- Could be spoofed

4. Attack on Analytics Assistant

4.1 Authorization

****Vulnerability:**** Scoped connection is defense in depth

An adversary can argue:

- SQL validation happens before connection
- Two layers could have different bugs
- The separation isn't proven

4.2 Test Provider

****Vulnerability:**** Deterministic is not representative

An adversary can argue:

- Templates don't reflect real LLM behavior
- The test doesn't validate real-world use
- Security holes could exist in real provider

4.3 Question Coverage

****Vulnerability:**** Template-based generation is limited

An adversary can argue:

- Only pre-defined patterns work
- Real questions would fail
- Coverage is artificial

4.4 Zero Violations Claim

****Vulnerability:**** Test set is not the sealed set

An adversary can argue:

- Only tested questions were checked
- The real sealed set could have violations
- Zero is not proof of security

5. Attack on Study Design

5.1 Target Trial Framework

****Vulnerability:**** Not actually a trial

An adversary can argue:

- No randomization occurred
- Observational data cannot emulate a trial
- The framework is aspirational only

5.2 Assumptions

****Vulnerability:**** All assumptions are untestable

An adversary can argue:

- No positivity verification
- Parallel trends untestable
- Exchangeability assumptions are asserted, not verified

5.3 Small Clusters

****Vulnerability:**** 8 components is too few

An adversary can argue:

- 4 per site is insufficient

- Standard errors are unreliable
- The analysis is underpowered

6. Summary of Attacks

7. Recommendations

1. **For Claims:** Require independent validation before making causal claims.
2. **For Services:** Add more adversarial tests, especially for the real LLM provider.
3. **For Design:** Either run an actual RCT or stop claiming causal effects.

This adversarial analysis is conducted in good faith to strengthen the overall assessment.